AP Computer Science

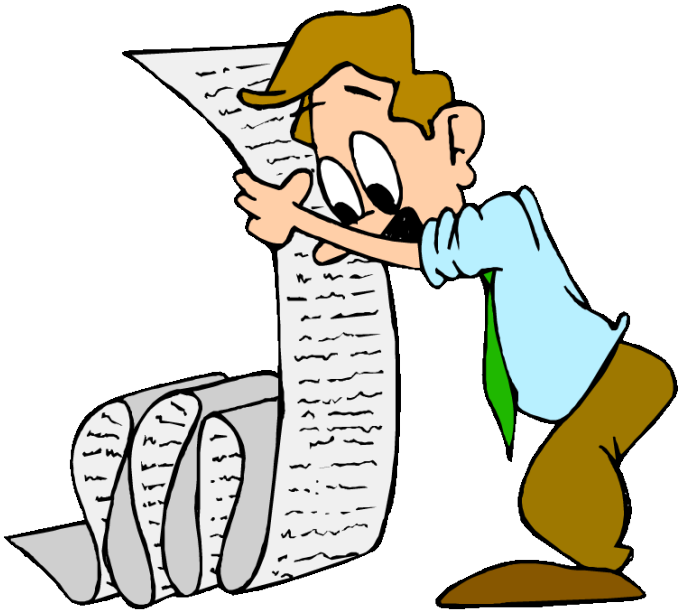
mr Hanley

Assignment 11/1011_2/13_8/B_{16} Version: 3.5

Binary	Ones Complement	Twos Complement

Last Updated: 3/26/2019 12:52 PM

List Manager



Create your own list management program

Lists, lists, lists!

Playlists, friend lists, to do lists, password lists, they go on and on and on…

Fortunately, now that you are learning the art of software development, you are going to create a list manager! also knows as a CRUD (create, record, update, delete system)

Think of some sort of entity that you are going to keep track of in a list.

Examples include;

* Dynamic Memory *Big O Notation*Stacks *Extreme Programming*Selection Sort*Insertion Sort*Waterfall Model

String*Arrays*ArrayList*Client Server*Artificial Intelligence*Inheritance*Files*Video Games*Short circuit evaluation*

Colleges you are applying to, friend's name and phone numbers and addresses, concerts attended, local restaurants, encryption methodologies, U2 songs, ski mountains, to do list, etc.

1. **Develop a class that implements the Comparable interface**

    a. This class must have at least 5 fields or data elements
    You may have only 3 Strings, other 2 FIELDS MUST BE DIFFERENT!!!

    b. While it is good OO design to have the variables private and the methods public, you may allow public variables for convenience

    c. Use good variable and method names (Charles, HiKC is not a good variable name!!!

2. Develop another class that allows you to manage a list of your first class. This will have an ArrayList inside it and have methods that will;

    a. **Load the Text File**
    Read in data from a text file into the ArrayList, making an object for each "record" read in and copying each "field" into that record (this should happen each time the program starts)

    b. **Display the list**
    There are 2 ways to display the list:
        i. Print each record on its own line.
            1. Use the \t or the System.out.format command so the columns line up
            2. Put a number to the left of each record so you can refer to them individually when you edit or delete!
            3. Put this is a method so you can call it from edit or delete to select which one to choose
            4. DO NOT JUST USE A METHOD THAT PRINTS THE FIELDS SEPARATED BY A PIPE!!!!
            5. Have a heading for each field and print the actual data on individual lines underneath the heading
            6. I like to have the name of the author and the number of records loaded and the data changed boolean printed in my display

        ii. Print each record on multiple lines
            1. See the run through of Jon StreetFighter's character manager on my YouTube Channel

    c. Sort the list according to the compareTo method or multiple sorts using Comparator objects (May use Collections.sort here)

        i. There are **two** different approaches to sorting
            1. Use a compareTo for the records and sort based on some sort of complexity.
                a. For a restaurant, this could mean computing a score for the restaurant based on food rating, cost and consistency

    d. Edit a record in the list (Update the fields)
    e. Delete a record in the list

f. Save the list back to the text file **(should prompt user if they attempt to exit program without saving)**

g. Archive the list to a separate file name for safety (when working with files, it is ALWAYS a good idea to save a backup)

There are lots of different file formats you can use when writing your ArrayList to the text file

My preferred version is as follows;

```
import java.io.*;
import java.util.StringTokenizer;  //allows breaking  a String into fields

/*File could look like this
 McCarthy|Walter|255 Grapevine Rd|Wenham|MA|01984|12000.00
 NaSmith|Courtney|7 Main St.|Clifton Park|NY|12065|18000.00
 Anderson|Trinity|957 First St.|Hermosa Beach|CA|01954|19000.00
 */
Need to hit enter after last record in file so cursor sits on next line

//You read in one line of the file which represents a record, divided by a
//symbol known as the delimiter, in this case the pipe |

BufferedReader input = new BufferedReader(new FileReader("data.txt"));

    String line;
    //Attempt to read from the file, prime the pump
    line = input.readLine();
    while (line != null) {//goes to the end of file
      StringTokenizer st = new StringTokenizer(line, "|"); //| is the delimiter
      //Now break up the line
      lname = st.nextToken();
      fname = st.nextToken();
      streetAddr = st.nextToken();
      town = st.nextToken();
      state = st.nextToken();
      zip = st.nextToken();
      salary = Double.parseDouble(st.nextToken());
```

```
        System.out.println("Here's our info " + fname + " " + lname + " " +
                    streetAddr + " " + town + " " + state + " " + zip +
                    " " + salary);  //obviously you must make and add to list!!!!
        line = input.readLine();  //must be at end so when no data while condition
trips
    }
    input.close();
//That code MUST be surrounded by a try catch block
```

NOTE:Your menu must include your name and the type of list that you are managing.

Example:

Welcome to the Ski Mountain List Database by mr Hanley

---

1 = Display Mountains
2 = Load in from Disk File
3 = Add a new Mountain
4 = Edit an existing Mountain
5 = Remove a Mountain
6 = Sort the List
7 = Save the List Back to Disk File
8 = Exit this program

When someone chooses, delete or edit, MAKE SURE they know what to type, Many of the programs I have run are very UNCLEAR.

For example, in this case,

5

Which mountain to delete?

1 = Killington
2 = Mount Snow
3 = Gore Mtn.
4 = Sunday River Resort

3

Removing Gore Mtn!

1

#	Name	Trails	Cost	Distance	Rating
1	Killington	121	92	121 miles	8.9
2	Mount Snow	75	78	101 miles	8.1
3	Sunday River Res	89	88	211 miles	8.5

MAKE SURE PEOPLE CAN SAVE THEIR ARRAYLISTS FROM THE MENU WHEN THEY WANT TO MAKE A SELECTION. READING IN CAN BE DONE IN THE BEGINNING OF THE PROGRAM ONLY OR AS A MENU OPTION

3. Rakowsky Factor:
   For those of you using the Scanner to read in via nextLine and then nextInt or nextDouble, see the following code example:

```
while(inFile.hasNextLine()) {
   String tempName = inFile.nextLine();
   int tempAge = inFile.nextInt();

   …more reads
   if(inFile.hasNextLine()){
      inFile.nextLine();  //Skip the blank that the Scanner will choke on!
   }
}
```

4. You may use Swing or console for this project

5. BONUS: Encrypt your data using Blowfish encryption

Project Name	**Assign 11 – List Manager**
Class 1 Name	Restaurant.java (Example)
Class 2 Name	RestManager.java (Example)
Class 3 Name	Possible Comparator Classes (Optional)
Text File1 Name(Needs to be in project folder NOT src or classes)	Restaurants.txt
Text File2 Name(Needs to be in project folder NOT src or classes)	Restaurants.bak

Restaurants.txt text file stores information about restaurants
COULD look like this;
Forno Bistro|Italian|7.5|Moderate|Chic Atmosphere|Saratoga Springs
Bellinis|Italian|8|Moderate|Good Atmosphere|Clifton Park
….
read in at beginning of program into arraylist

write out at end of program in order to allow data to persist = persistency
write out a copy just in case to a separate file (files can become corrupted)

FoodGenre.java

```
enum FoodGenre {Italian,Chinese, American, Indian, BarbQue};
```

Restaurant.java

```
public class Restaurant implements Comparable{
	public String name;
	public FoodGenre gen;
	//etc

	public int compareTo(Object o) {  //Fixed 1/15/2016
		Restaurant other = (Restaurant) o;  //type cast so we can compare
		…

	}
}
```

RestManager.java

```
public static ArrayList<Restaurant> list = new ArrayList<Restaurant>();

public static void main() {

	//Has a menu which allows control of ArrayList
}
```

Remember to READ in all of your elements

Rubric	
Entity Class	
-compareTo	20
-good variable names with at least 5 data fields	10
main class or Frame for GUI	0
-menu and options are user friendly and easy to follow	15
	0
-reads from text file	15
-sorts based on compareTo	15
-edits	15
-adds	15
-deletes	15
-displays	15 -7 if you just use fields separated by pipes Use System.out.format with codes for field widths
-saves	15
-archives(must be a separate file)	10
-warns user if attempting to exit program without saving unsaved changes	5
-comments and variable names	10
TOTAL	175
BONUS	20

*Recursion*Linear Search*Binary Search*Grid World Case Study*File Processing *nlogn*Hangman*

How to Turn in Your CRUD

For this application, you are going to create a WORD document with an entire run through of your program…..YOUR FILE NAME MUST BE CALLED **Crud_Output_Jon_Street**.(Unless you are not Jon Street)

Make sure you have at least 4 records saved to your disk file.

Your word document will look like this…..

Name: Jon Street AKA StreetFighter. Period: 7 APCS APCS Java CRUD RunThrough

<u>Purpose:</u>
This program's purpose is to allow the user to manage a list of characters from an adventure game.
Each character has stats that are kept track of;
name, class, gender, level (start at level 1 and go up) and experience (0-99).
Once a character reaches 100 experience points, they go up to the next level and experience goes back to 0.

Characters will be loaded in from the local file system.
The user can display them, edit any of the fields in them, add new characters and remove characters from the list.
Saving to disk and reading in from disk as well as archiving to a backup is supported.

<u>compareTo</u>
My compareTo method allows two characters to be compared mathematically.
The way is works is that levels are compared.
Higher levels will get moved to the front of the ArrayList when the Collections.sort logic is executed.
If the levels are the same, the compareTo resorts to comparing names alphabetically by ASCII code.

My demonstration below will show that newly added characters add by default at the end but that when Collections.sort executes, characters with higher levels will move to the top.

Test Sequence for CRUD Application		
1. Open your Application		
2. Read in text file	MUST HAVE 4 records already out there	
3. Display	See the 4 records	
4. Delete an item	Need to present the user with a list so they can see what to choose. I prefer to see a message like "Jeff has been deleted."	
5. Display	One less should be there.	
6. Edit an item	Allow user to choose which field to edit	
7. Display	Should show altered record.	
8. Add a new item	Type in new data MAKE SURE IT BELONGS HIGHER UP WHEN SORTED (don't make it just garbage input) Add at end	
9. Display	Show new record at bottom	
10. Sort	Make sure the order is CHANGED after sort (I prefer some sort of message like "**List is now sorted by level then alphabetical by name**"	
11. Optional 2nd Sort If you used a Comparator and sort on a separate criteria	Run your next sort and then display your output, add a comment in your word doc describing the new order of the list.	
12. Display	Should show updated positions.	
13. Save	Either save to both normal file and archive at once or make separate menu options.	
14. Optional Separate Archive		
15. Display		
16. Change a record somehow (either delete, add a new or edit a record)		
17. Try to quit the program	The program should warn you that you are about to lose unsaved data…choose that you want to save	
18. Exit java program		
19. Restart and Read in		
20. Display	Should have records that were updated as of the last run of the program.	

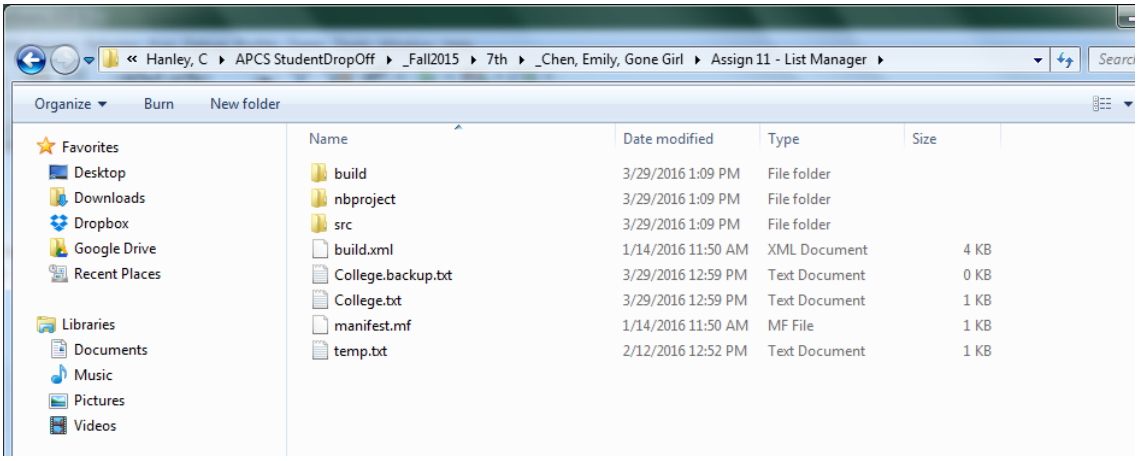
1st Run Through of CRUD
(must be red in font size 28 font)

<Perform Steps 1-18 and paste into word, each section must be preceded by a red font 28 heading>

2nd Run Through of CRUD

<Perform Steps 19-20 and paste into word>

Take a screen shot of your directory in your project showing your .txt file and backup file names
Hit printscreen and paste it into MS Word so I can see both file names




```
//Copy Some of your comments to earn your 10 points for comments at the bottom
String result = "";
// this will reference one line at a time
String line = null;
try {
   // FileReader reads text files in the default encoding.
   FileReader fileReader = new FileReader(fileName);
   // always wrap FileReader in BufferedReader.
   BufferedReader bufferedReader = new BufferedReader(fileReader);
   line = bufferedReader.readLine();
   while (line != null) {
      result += line + "\n";
      line = bufferedReader.readLine();
   }
   // always close files.
   bufferedReader.close();
   // exception handling
```

Paste in your compareTo method in here: (or each of your Comparator classes)

```
public int compareTo(Object o) { //Getting an object to compare said object to

   Character other = (Character) o; //Casting the character

   //First check levels....
   if(level == other.level){
      //if levels same, then go to name as secondary ordering
      return name.compareTo(other.getCharacterName());
   }
```

```
    else return level - other.level; //Checking to see which level is greater and if it needs sorted
}
```